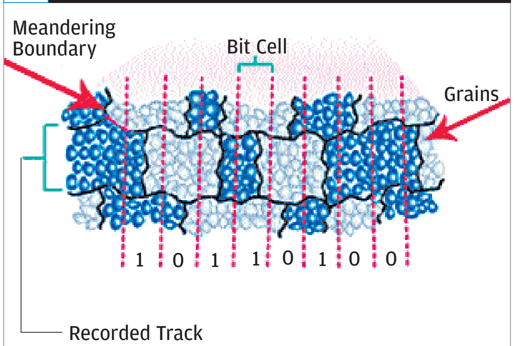
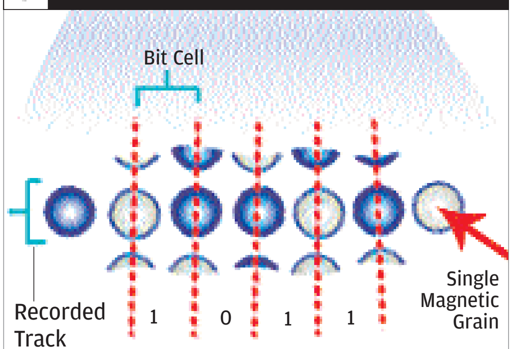
**In patterned media, we need just about one grain-sized volume per bit. Density can therefore be increased by about two orders of magnitude**

### Conventional Media




Look along the recorded track. A region of change in magnetisation of the grains indicates a 1, and a region of no change indicates a 0

### Patterned Media




Here, a single magnetic grain can hold a bit. Along the recorded track, a change in grain magnetisation means a 1, and no change means a 0

## Patterned Media

Patterned media is looking considerably into the future. However, we mention it because it is an important technology being looked into, and might well be the distant future of hard disks. Seagate's Mark Kryder believes PMR might take areal densities to as much as 1 terabit psi, but doesn't expect it to go any further than that. On the other hand, he points out that work is continuing on other technologies that can be ready when the industry needs them. HAMR and patterned media recording are the two most favoured alternatives.

John Best, CTO of Hitachi GST, has said that PMR "in its first two generations probably provides a fairly modest areal density increase, on the order of a factor of two or three." Bigger advances will come when PMR is linked to a new generation of so-called patterned media, Best said.

The idea of patterned media is this: like we said before, a few hundred magnetic grains are required to store a bit. Densities could increase drastically if we could find a way to store one bit per grain. This can happen only when the grains are ordered in a particular way—that's why it's called patterned media. Of course, this assumes that drive makers can find economical ways to read and write the patterns.

"I'm a bit of a patterned-media enthusiast myself, but I couldn't predict the time frame it might happen," Best said. "It might come before, after or at the same time as work in HAMR."

So here's how patterned media works: look at the figure alongside. A '0' is indicated when there is no transition in magnetisation, and a '1' is indicated when there is a transition. The boundaries between regions of opposite magnetisation must occur along the boundaries between the grains—this makes the boundaries meandering, approximating the ideal straight boundary. Remember that the straighter the boundary is, the easier it is to detect (read).

If the grains are small enough, the magnetic transitions are straight enough, so it's easy to detect which bit cells contain a boundary and which do not. However, if the density is increased (meaning the bit cells are shrunk) without shrinking the grains, the magnetic transitions become noisier, eventually preventing the head from accurately reading the data. To keep the noise associated with grain boundaries small enough for reliable data detection is why fifty to hundred grains are needed per bit cell.

Now, the solution to achieve higher densities is obvious: make smaller grains. As of now, however, grain sizes have gotten so small that further shrinkage would bring in the superparamagnetic effect. But with patterned media, each bit is stored in a single, deliberately-formed magnetic volume. This may be one grain or several coupled grains, rather than a collection of random decoupled grains. The magnetic transitions no longer meander between random grains, but form perfectly distinct boundaries between precisely located islands. This means that we need just about one grain-sized volume per bit. Density can therefore be increased by about two orders of magnitude.

## What To Look Out For

AFC media will be here for a while, and PMR will take off in a big way this year. On the whole, the picture looks perfect enough. This is one industry that has everything going for it. There's a fantastically huge market, which will only grow; so far, the roadblocks have been cleared, and research is already underway to clear the hurdles ahead. Prepare for 1,000 GB hard disk within 12 months from now—and prepare for a 2,000 GB hard disk in the year after that!

The only change that you will probably see is hard disks being used in more and more devices. Expect to see hard disks in places you haven't seen them before. Also, as hard disks become smaller and smaller and more affordable, they will come more directly in competition with the other storage media, such as Flash and MRAM.

Some aver that holographic storage is the future of storage-refer the March 2004 issue of *Digit*. It may be, but like we said earlier, practical, everyday holographic storage is still very much in the future, and the hard disk industry need not worry about it yet.

The immediate future, beginning this year, is PMR—just remember that you read about it here first! ■

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